

Name:

Collaborators:

**Instructions:** Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

## Determine Equilibrium Stability

### 1. Equilibriums and Their Stability

In a 1st-order autonomous ODE  $\frac{dy}{dt} = f(y)$ , an equilibrium occurs at a point  $y^*$  where  $f(y^*) = 0$ . The *stability* of the equilibrium  $y^*$  depends on how nearby solutions behave:

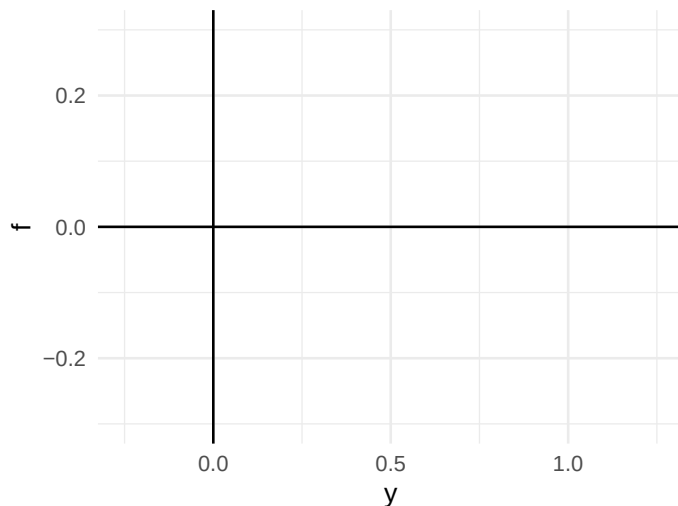
- If solutions starting close to  $y^*$  move toward it in time, the equilibrium is *stable*.
- If solutions starting close to  $y^*$  move away from it in time, the equilibrium is *unstable*.

Consider the ODE,  $\frac{dy}{dt} = y - y^2$ .

a. Define  $f(y)$ . Then, determine all equilibrium solutions,  $y(t) = y^*$ .

b. Compute  $\frac{\partial f}{\partial y}$  evaluated for all  $y^*$ . Then, mark each if positive or negative.

c. Use the blank axis below to complete the Phase Diagram of the given ODE and mark each equilibrium whether they are stable or unstable.



d. What do you observe about the stability of the equilibriums and the signs of the partial derivatives computed in Part (b)?

## 2. Semi-Stable Equilibriums

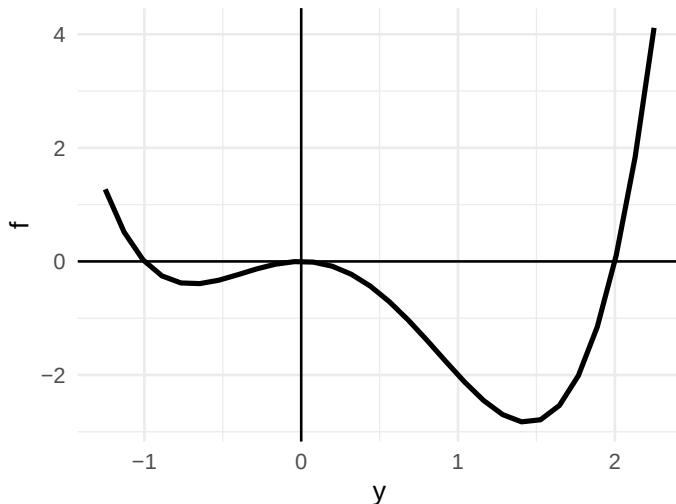
If solutions approach  $y^*$  from one side but move away from it on the other side, the equilibrium is *semi-stable*.

Consider the ODE,  $\frac{dy}{dt} = (y - 2)(y + 1)y^2$ .

- a. Define  $f(y)$ . Then, determine all equilibrium solutions,  $y(t) = y^*$ .

- b. Compute  $\frac{\partial f}{\partial y}$  evaluated for all  $y^*$ . Then, mark each if positive or negative.

- c. Complete the Phase Diagram of the given ODE shown below and mark each equilibrium whether they are stable or unstable.



- d. What do you observe about the stability of the equilibriums and the signs of the partial derivatives computed in Part (b)?

- e. Use the phase diagram to describe the behavior of solutions –including where it approaches– given the following conditions. Explain your reasoning.

i.  $-1 < y(0) < 0$

ii.  $0 < y(0) < 2$

- f. Show using  $\frac{\partial^2 f}{\partial y^2}$  evaluated at a semi-stable equilibrium that the solutions around this equilibrium is either decreasing or increasing.